NAME: KEDAR JOSHI

REPORT FOR RESEARCH CHALLENGE (TOWER LAB, UTSW)

DATE : 1/22/2026

CONTENTS

1. Abstract

2. Materials and Methods
   2.1 : Datasets
   2.2 : Data Processing

3. Results

4. Conclusion

5. References

6. Professional References

1. Abstract

This study investigates how neutrophil–macrophage communication varies across three anatomical compartments: bone marrow (BM), blood, and injured tissue. Using single-cell RNA sequencing data and the NicheNet framework, I inferred ligand–target interactions from neutrophils to macrophages within each compartment. By integrating cell-type-specific expression, ligand–receptor knowledge, and downstream macrophage gene programs, we identified compartment-specific and conserved signaling axes. My analysis reveals a shared core of ligands involved in macrophage maintenance and lipid handling, alongside distinct compartment-specific signaling programs reflecting homeostasis, systemic inflammation, and tissue repair.

2. Materials and Methods

2.1: Datasets

- Bone Marrow (BM): scRNA-seq dataset GSM6845846 (10X Genomics)
- Peripheral Blood (Blood): scRNA-seq dataset GSM6845847 (10X Genomics)
- Injury (HO): scRNA-seq dataset GSM6845848 (10X Genomics)

2.2: Data Processing

Single-cell RNA-seq datasets from bone marrow, peripheral blood, and injury-associated tissues were processed independently using Seurat. Cell identities were assigned based on curated cluster annotations, and neutrophil and macrophage populations were subset for downstream analysis. Differential expression analysis was used to define macrophage receiver gene sets within each compartment.

Ligand–target inference was performed using the NicheNet framework. Expressed ligands were defined as genes detected in neutrophils, while receiver gene sets were derived from macrophage marker genes filtered to those expressed in macrophages. Ligand activities were scored using Pearson correlation between predicted and observed target gene expression. The top-ranked ligands per compartment were retained for comparative analysis.

3. Results

This study provides a compartment-resolved analysis of neutrophil–macrophage communication across the bone marrow, peripheral blood, and injured tissue using ligand prioritization via NicheNet. While a small core of ligands most notably APOE, CSF1, and SPP1 was conserved across compartments, their relative ranking and accompanying signaling programs differed markedly, indicating strong contextual rewiring of neutrophil-driven macrophage instruction.

In the bone marrow, neutrophil–macrophage signaling was dominated by ligands associated with immune homeostasis, cellular retention, and myeloid maturation. The prominence of CSF1 supports a role for neutrophils in sustaining macrophage survival and differentiation within the hematopoietic niche. Adhesion and trafficking molecules such as SELP and SELPLG suggest physical and signaling interactions that may regulate macrophage localization and readiness. Additional enrichment of IL15 and HLA-DMA points toward immune priming and antigen-presentation-related processes rather than overt inflammatory activation, consistent with the bone marrow's role as a controlled developmental environment.

In contrast, peripheral blood exhibited a shift toward inflammatory and recruitment-driven communication. High-ranking ligands including CCL2, TNF, and IL1B are canonical drivers of monocyte and macrophage chemotaxis, activation, and cytokine amplification. This pattern aligns with neutrophils acting as early systemic sentinels that rapidly alert and mobilize macrophages during immune surveillance. The concurrent presence of TGFB1 suggests that regulatory or dampening signals may be engaged in parallel, potentially limiting excessive inflammation in circulation.

At the injury site, neutrophil–macrophage signaling was most strongly reprogrammed toward tissue repair, angiogenesis, and macrophage polarization. The dominance of IL4 supports neutrophil-driven skewing of macrophages toward alternatively activated pro-repair phenotypes. Strong enrichment of VEGFA, ANGPT2, PDGFA, and ICAM1 indicates active coordination of vascular remodeling, endothelial interaction, and tissue regeneration. Notably, APOE emerged as the top-ranked ligand in this compartment, implicating lipid handling, efferocytosis, and resolution of inflammation as central components of neutrophil–

macrophage crosstalk during healing.

Collectively, these findings reveal neutrophils as highly adaptive signaling hubs whose influence on macrophages is profoundly shaped by anatomical context. Rather than deploying a fixed communication program, neutrophils modulate macrophage survival, activation, and functional polarization in a compartment-specific manner, supporting niche maintenance in bone marrow, immune activation in blood, and repair-oriented responses at sites of injury. This provides a mechanistic framework for understanding inflammatory progression and resolution and highlights potential context-specific therapeutic targets for immune modulation in disease and tissue repair.

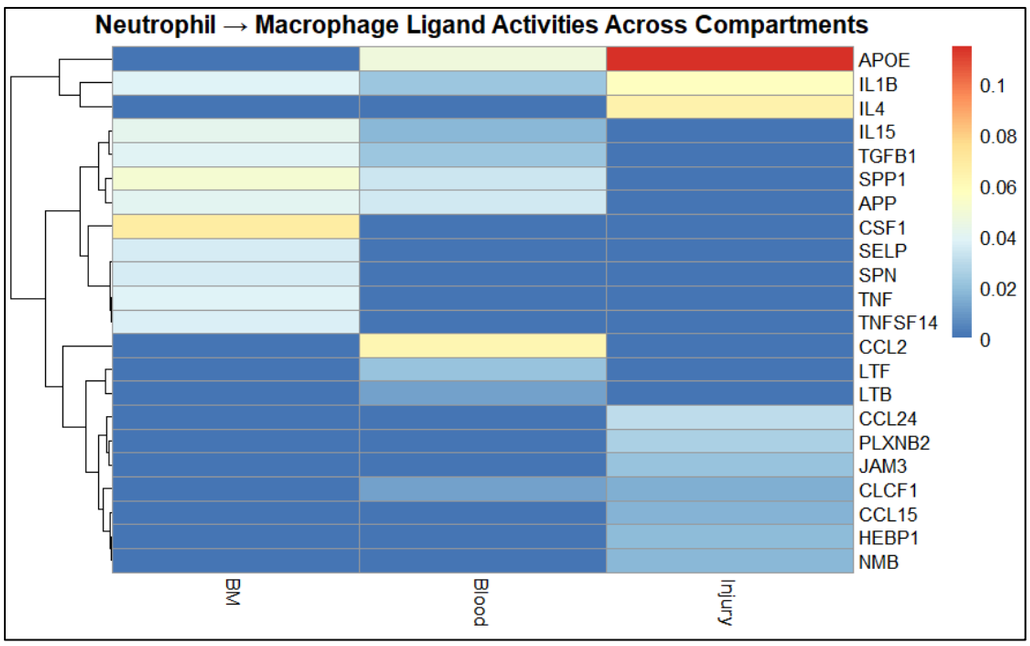

Fig1. Pheatmap of ligand activity across compartments

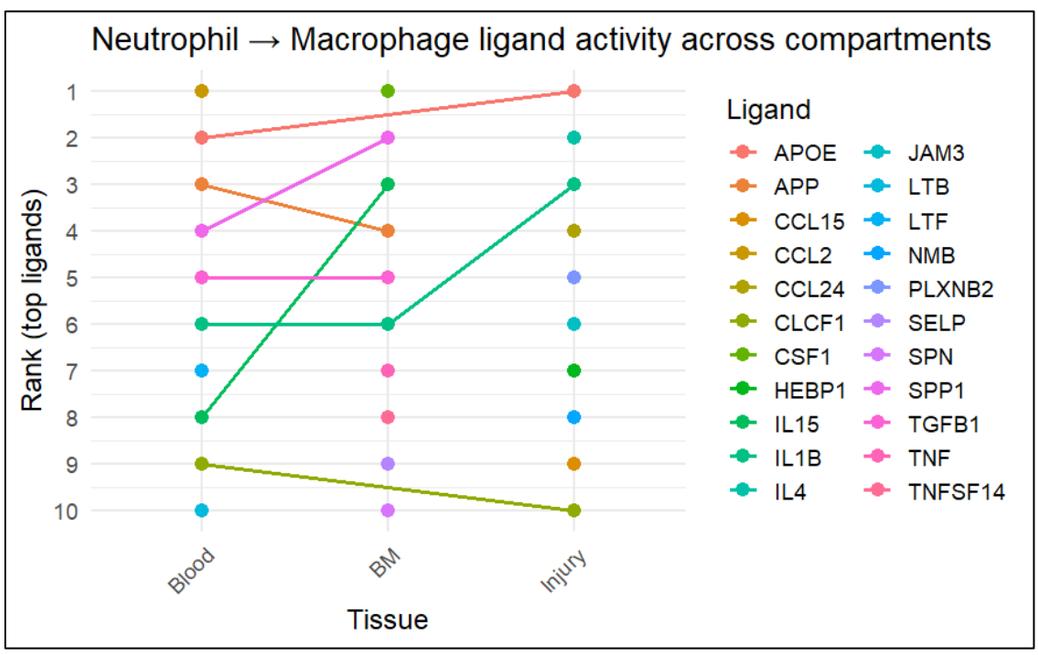

Fig2. Ligand activity across compartments

4. Conclusion

Together, these findings demonstrate that neutrophil–macrophage communication is highly context dependent. While a conserved core of ligands supports macrophage viability across tissues, compartment-specific ligands reflect the functional demands of each environment. Bone marrow signaling emphasizes immune homeostasis, blood signaling promotes systemic inflammation and recruitment, and injury-site signaling drives macrophage polarization and tissue repair.

References

1. Mantovani, A., Cassatella, M. A., Costantini, C., & Jaillon, S. (2011). *Neutrophils in the activation and regulation of innate and adaptive immunity. Nature Reviews Immunology.* https://pmc.ncbi.nlm.nih.gov/articles/PMC10619500/
2. Browaeys, R., Saelens, W., & Saeys, Y. (2020). *NicheNet: modeling intercellular communication by linking ligands to target genes. Nature Methods, 17*, 159–162.
3. Stuart, T., et al. (2019). *Comprehensive integration of single-cell data. Cell, 177*(7), 1888–1902.e21.
4. Kolaczkowska, E., & Kubes, P. (2013). *Neutrophil recruitment and function in health and inflammation. Nature Reviews Immunology, 13*(3), 159–175.
5. Murray, P. J., & Wynn, T. A. (2011). *Protective and pathogenic functions of macrophage subsets. Nature Reviews Immunology, 11*(11), 723–737.
6. Williams, A. E., & Jose, P. J. (2019). *Role of neutrophils in resolution of inflammation. British Journal of Pharmacology, 176*(22), 3781–3792.
7. Herrero, J., et al. (2016). *Ensembl comparative genomics resources. Database (Oxford).* https://www.ensembl.org/biomart

Professional References:

1. **Clarence Wang, Vice President Data Science, Dyne Therapeutics (cwang@dyne-tx.com)**
2. **Vinod Yadav, Associate Director Data Science, Dyne Therapeutics (vyadav@dyne-tx.com)**
3. **Evan Fernandez, Senior Bioinformatician, Weill Cornell Medicine (emf2009@med.cornell.edu)**